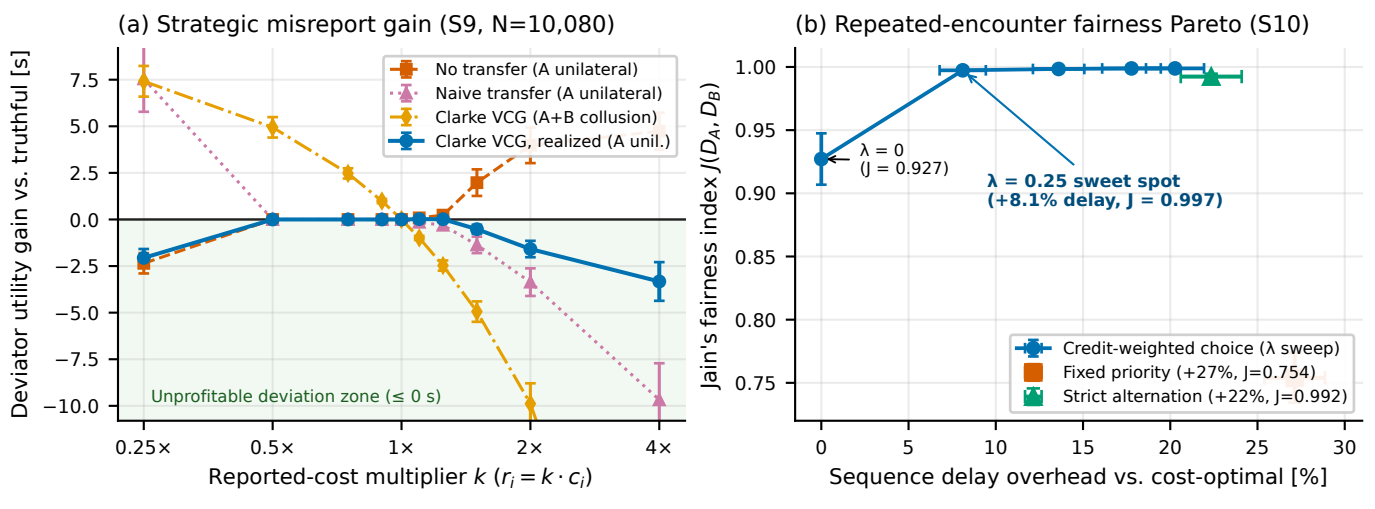
(a) Strategic misreport gain (S9, N=10,080)
No transfer (A unilateral)
Naive transfer (A unilateral)
Clarke VCG (A+B collusion)
Clarke VCG, realized (A unil.)
Deviator utility gain vs. truthful [s]
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
Unprofitable deviation zone (≤ 0 s)
0.25×
0.5×
1×
2×
4×
Reported-cost multiplier k ($r_i = k \cdot c_i$)
(b) Repeated-encounter fairness Pareto (S10)
Jain's fairness index $J(D_A, D_B)$
1.00
0.95
0.90
0.85
0.80
0.75
λ = 0
(J = 0.927)
λ = 0.25 sweet spot
(+8.1% delay, J = 0.997)
Credit-weighted choice (λ sweep)
Fixed priority (+27%, J=0.754)
Strict alternation (+22%, J=0.992)
0
5
10
15
20
25
30
Sequence delay overhead vs. cost-optimal [%]